

Class 11: AlphaFold

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Section 7: Interpreting Results

After running the HIV protease dimer sequence in the AlphaFold MMSeqs2 notebook (with default settings) and acquiring a zip file, we can begin result interpretation.

Visualization of the models and their estimated reliability

We will use Mol* to visualize the model PDB files for the HIV protease dimer.

Step 1: Loading the five PDB files into Mol*.



Figure 1: The five PDB dimer structures, when first loaded into Mol*, are not in the same reference frame.

Step 2: Performing superposition of the five models. A single chain is aligned at a time.



Figure 2: Superposed dimer models.

Step 3: Coloring our superposed dimer model by Mol* Uncertainty/Disorder (i.e. using the pLDDT scores derived from AlphaFold)

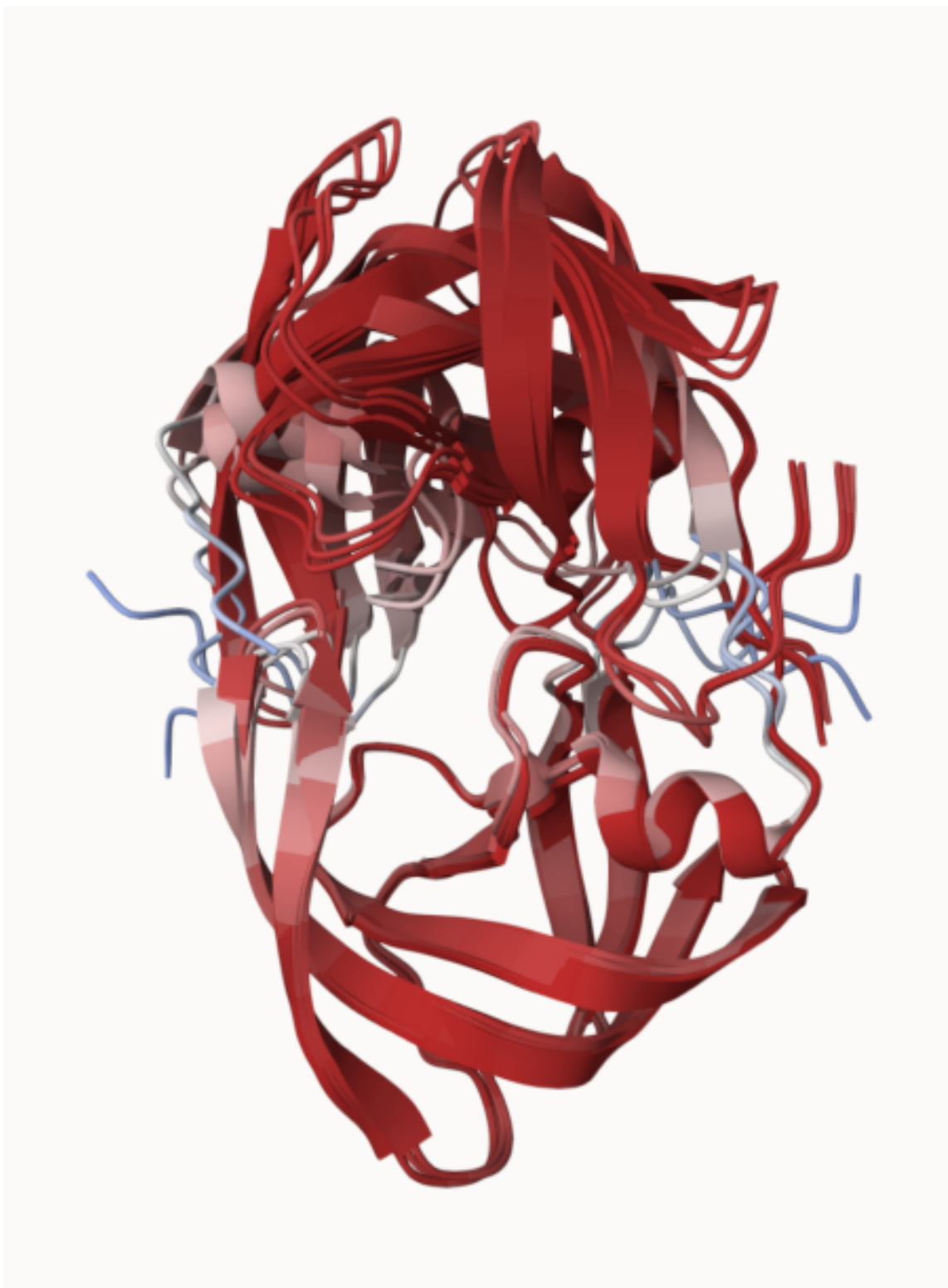


Figure 3: pLDDT colored HIV-Pr dimer structure models. Red indicates high confidence, blue indicates low confidence.

We see that secondary structures (i.e. alpha helices and beta sheets) are generally colored pink/red and have higher confidence scores, whereas non-compact loop regions (which may be intrinsically disordered regions, or IDRs) are more likely to be white/blue and have lower confidence scores. Furthermore, by browsing each chain in Residue-selection mode, it seems regions at the N and C termini are more likely to have lower confidence scores.

Step 4: Exploring other coloring settings. For instance, we can color residues by hydrophobicity.



Figure 4: Hydrophobicity colored HIV-Pr dimer structure models. Hydrophobic/nonpolar residues are colored green, and hydrophilic/polar residues are colored orange/red.

Coloring the residues by hydrophobicity shows that the HIV protease dimer appears to have a more hydrophobic (green) core. The dimer's hydrophilic residues (orange/red) are generally positioned along the surface of the protein. This suggests that the protease may be a soluble protein (i.e. found in the cytosol, as opposed to being an integral membrane protein).

Section 8: Custom analysis of resulting models

We need to read in the data from our AlphaFold models.

```
library(bio3d)

# Method 1: Use read.pdb() to read in an individual PDB file.
pdb <- read.pdb("hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer.pdb")
```

```
Call: read.pdb(file = "hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer.pdb")
```

```
Total Models#: 1
  Total Atoms#: 1514, XYZs#: 4542 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 0 (residues: 0)
Non-protein/nucleic resid values: [ none ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, calpha, call
```

Alternatively, make a vector of input PDB file names that we can read into R:

```
# Method 2: First make vector of file names for all PDB models.
pdbfiles <- list.files(path = "hivpr_dimer_23119/",
  pattern = ".pdb",
  full.names = TRUE)
```



```
# Print PDB file names.
pdbfiles
```

```
[1] "hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0"
[2] "hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0"
[3] "hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0"
[4] "hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0"
[5] "hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0"
```

```
# Read all data from models using pdbaln() (from BioCon's msa package)
# Superpose coordinates using fit argument
pdbs <- pdbaln(pdbfiles, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
.....
```

Extracting sequences

```
pdb/seq: 1    name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0
pdb/seq: 2    name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0
pdb/seq: 3    name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
pdb/seq: 4    name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
pdb/seq: 5    name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
```

```
# Print PDB model sequences.
pdbs
```

```

1          .          .          .          .          50
[Truncated_Name:1]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:2]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:3]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:4]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:5]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
*****
1          .          .          .          .          50
```

```

51 . . . . 100
[Truncated_Name:1]hivpr_dime GGIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_dime GGIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_dime GGIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_dime GGIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_dime GGIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
*****
51 . . . . 100

101 . . . . 150
[Truncated_Name:1]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMKMIGGIG
[Truncated_Name:2]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMKMIGGIG
[Truncated_Name:3]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMKMIGGIG
[Truncated_Name:4]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMKMIGGIG
[Truncated_Name:5]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMKMIGGIG
*****
101 . . . . 150

151 . . . . 198
[Truncated_Name:1]hivpr_dime GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_dime GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_dime GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_dime GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_dime GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
151 . . . . 198

```

Call:

```
pdaln(files = pdbfiles, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD

Calculate the RMSD (measure of structural distance between coordinate sets) between all pairs of models.

```
# Calculate RMSD values, and store as an RMSD matrix
rd <- rmsd(pdb, fit=T)
```

Warning in rmsd(pdb, fit = T): No indices provided, using the 198 non NA positions

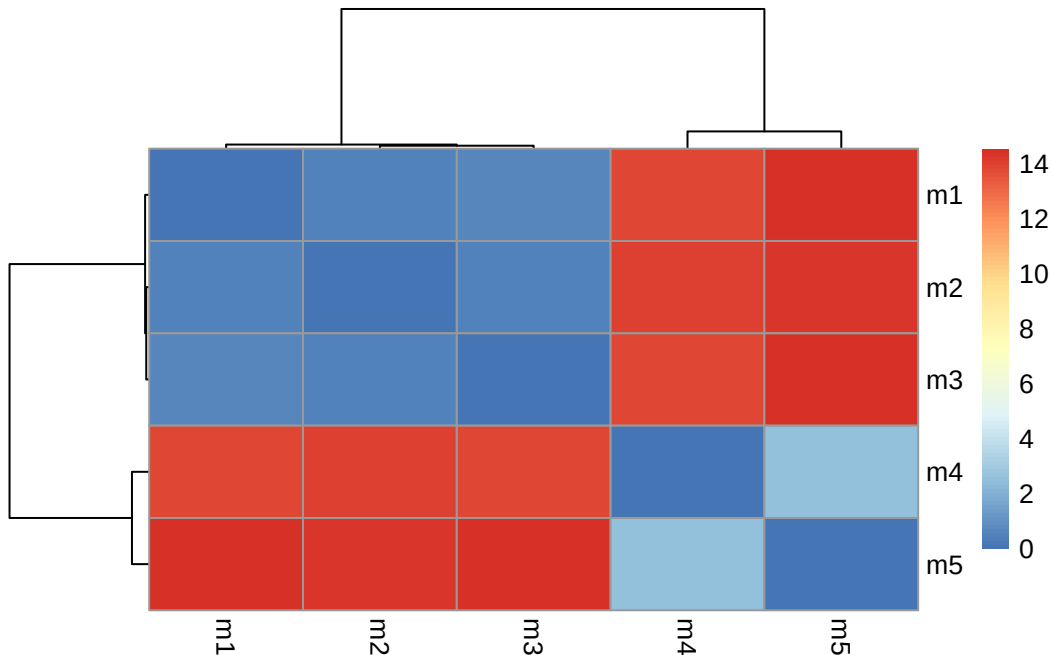
```
# Calculate range of RMSD scores
range(rd)
```

```
[1] 0.000 14.526
```

Make a heatmap of RMSD matrix values:

```
library(pheatmap)

# Rename columns and rows of RMSD matrix
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
# Generate heatmap from matrix
pheatmap(rd)
```



From my AlphaFold results, it appears that Models 2 and 3 are the most similar pair of models (have the smallest pairwise RMSD). Models 2 and 3 are in turn fairly similar to Model 1, and this trio of models is relatively *dissimilar* to models 4 and 5 (i.e. they separate into two distant clusters). Models 4 and 5 are similar to each other (though not as similar as models 1, 2, and 3 are to each other).

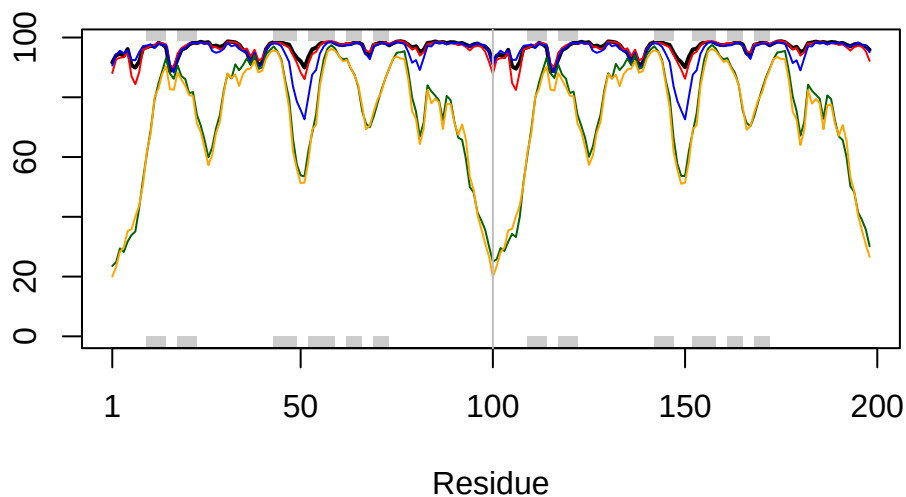
pLDDT

Plot the pLDDT values across all models.

```
# Read the reference PDB structure, to later plot its secondary structure data
pdb_ref <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
# Plot the pLDDT values for each model (stored in B-factor column, pdbb$b)
plotb3(pdbb$b[1,], typ="l", lwd=2, sse=pdb_ref)
points(pdbb$b[2,], typ="l", col="red")
points(pdbb$b[3,], typ="l", col="blue")
points(pdbb$b[4,], typ="l", col="darkgreen")
points(pdbb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



Using `core.find()` to superpose our models

To improve the superposition:

```
# Use core.find() to identify consistent core atom positions across all models:
core <- core.find(pdbbs)
```

```
core size 197 of 198  vol = 5437.294
core size 196 of 198  vol = 4705.336
core size 195 of 198  vol = 1827.704
core size 194 of 198  vol = 1121.539
core size 193 of 198  vol = 1047.76
core size 192 of 198  vol = 999.32
core size 191 of 198  vol = 953.718
core size 190 of 198  vol = 910.755
core size 189 of 198  vol = 870.203
core size 188 of 198  vol = 836.304
core size 187 of 198  vol = 805.237
core size 186 of 198  vol = 775.99
core size 185 of 198  vol = 752.564
core size 184 of 198  vol = 712.023
core size 183 of 198  vol = 685.568
```

core size 182 of 198	vol = 663.911
core size 181 of 198	vol = 645.881
core size 180 of 198	vol = 627.97
core size 179 of 198	vol = 611.812
core size 178 of 198	vol = 595.931
core size 177 of 198	vol = 581.132
core size 176 of 198	vol = 566.736
core size 175 of 198	vol = 548.587
core size 174 of 198	vol = 534.114
core size 173 of 198	vol = 505.214
core size 172 of 198	vol = 491.225
core size 171 of 198	vol = 473.905
core size 170 of 198	vol = 460.426
core size 169 of 198	vol = 444.81
core size 168 of 198	vol = 431.661
core size 167 of 198	vol = 421.542
core size 166 of 198	vol = 405.601
core size 165 of 198	vol = 392.666
core size 164 of 198	vol = 381.077
core size 163 of 198	vol = 367.559
core size 162 of 198	vol = 358.379
core size 161 of 198	vol = 346.865
core size 160 of 198	vol = 334.809
core size 159 of 198	vol = 324.09
core size 158 of 198	vol = 312.153
core size 157 of 198	vol = 301.296
core size 156 of 198	vol = 290.431
core size 155 of 198	vol = 281.319
core size 154 of 198	vol = 272.529
core size 153 of 198	vol = 263.215
core size 152 of 198	vol = 253.54
core size 151 of 198	vol = 240.86
core size 150 of 198	vol = 227.447
core size 149 of 198	vol = 215.581
core size 148 of 198	vol = 202.041
core size 147 of 198	vol = 195.426
core size 146 of 198	vol = 188.721
core size 145 of 198	vol = 181.778
core size 144 of 198	vol = 173.615
core size 143 of 198	vol = 165.946
core size 142 of 198	vol = 156.117
core size 141 of 198	vol = 149.814
core size 140 of 198	vol = 143.616

core size 139 of 198 vol = 135.81
core size 138 of 198 vol = 127.851
core size 137 of 198 vol = 122.596
core size 136 of 198 vol = 117.203
core size 135 of 198 vol = 109.848
core size 134 of 198 vol = 104.812
core size 133 of 198 vol = 98.776
core size 132 of 198 vol = 94.799
core size 131 of 198 vol = 90.494
core size 130 of 198 vol = 87.403
core size 129 of 198 vol = 83.558
core size 128 of 198 vol = 79.08
core size 127 of 198 vol = 75.056
core size 126 of 198 vol = 71.238
core size 125 of 198 vol = 67.735
core size 124 of 198 vol = 64.289
core size 123 of 198 vol = 61.381
core size 122 of 198 vol = 57.515
core size 121 of 198 vol = 53.254
core size 120 of 198 vol = 48.654
core size 119 of 198 vol = 45.832
core size 118 of 198 vol = 41.819
core size 117 of 198 vol = 38.71
core size 116 of 198 vol = 36.294
core size 115 of 198 vol = 33.386
core size 114 of 198 vol = 30.472
core size 113 of 198 vol = 27.786
core size 112 of 198 vol = 25.403
core size 111 of 198 vol = 22.827
core size 110 of 198 vol = 21.106
core size 109 of 198 vol = 19.327
core size 108 of 198 vol = 17.796
core size 107 of 198 vol = 16.235
core size 106 of 198 vol = 14.508
core size 105 of 198 vol = 12.969
core size 104 of 198 vol = 11.834
core size 103 of 198 vol = 11.185
core size 102 of 198 vol = 10.298
core size 101 of 198 vol = 8.898
core size 100 of 198 vol = 7.813
core size 99 of 198 vol = 6.074
core size 98 of 198 vol = 5.286
core size 97 of 198 vol = 4.43

```

core size 96 of 198  vol = 3.873
core size 95 of 198  vol = 3.321
core size 94 of 198  vol = 2.855
core size 93 of 198  vol = 2.293
core size 92 of 198  vol = 1.937
core size 91 of 198  vol = 1.631
core size 90 of 198  vol = 1.331
core size 89 of 198  vol = 0.957
core size 88 of 198  vol = 0.803
core size 87 of 198  vol = 0.647
core size 86 of 198  vol = 0.532
core size 85 of 198  vol = 0.444
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 86 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     7   7      1
2     9  49     41
3    52  95     44

```

```

# Write out fitted structures, and store superposed coordinates in a new directory
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")

```

Open superposed coordinates (in `corefit_structures` directory) in Mol*. Color by Uncertainty/Disorder (B-factor column with pLDDT scores).

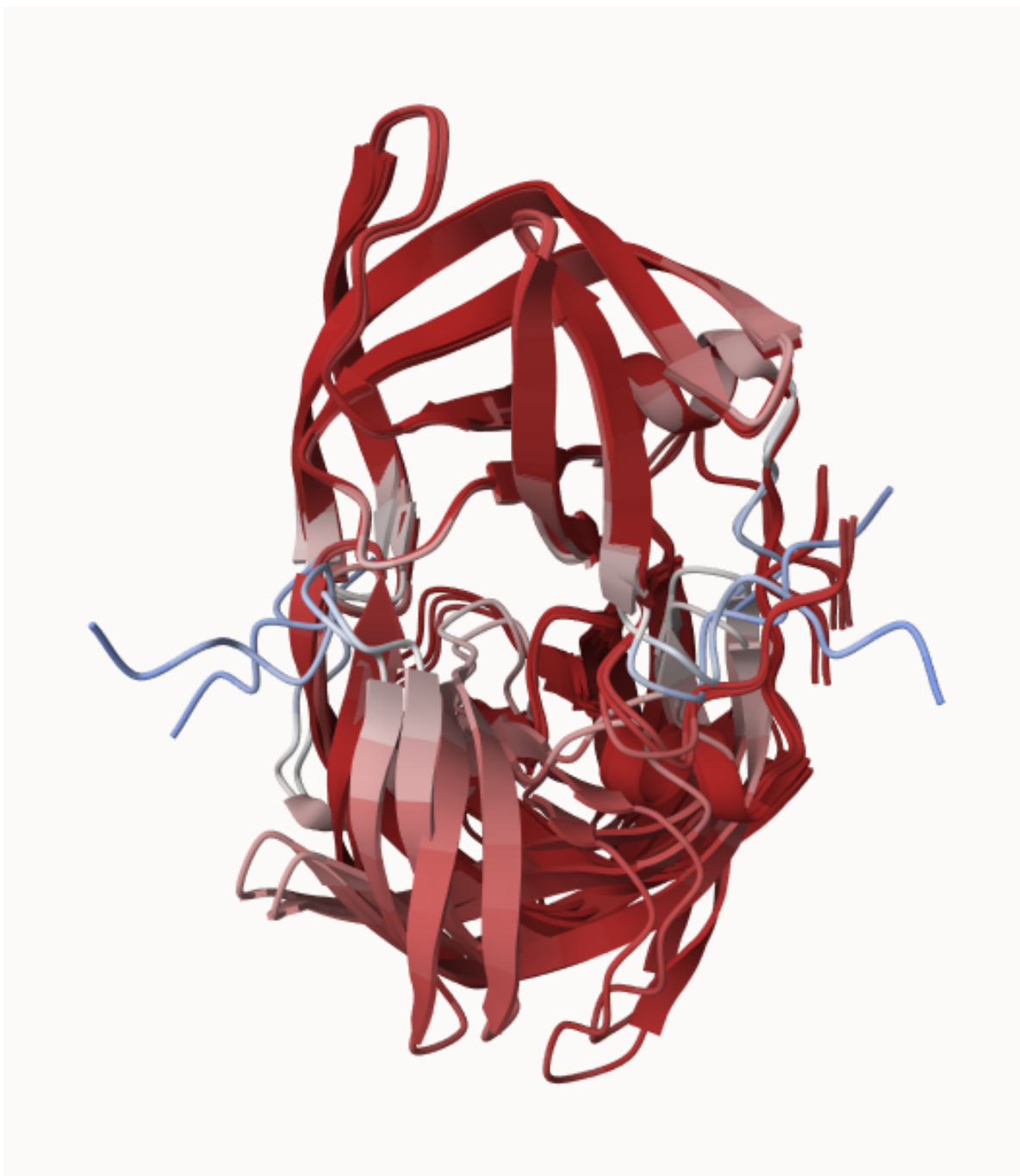


Figure 5: All core superposed structures in Mol*, colored by B-factor pLDDT scores.

One of the two chains (positioned at the bottom, in the orientation shown above) appears to be quite variable across the 5 models, whereas the other chain (positioned at the top, in the

orientation shown above) is fairly consistent.



Figure 6: Core superposed structures of Models 1, 2, & 3 ONLY in Mol*



Figure 7: Core superposed structures of Models 4 and 5 ONLY in Mol*

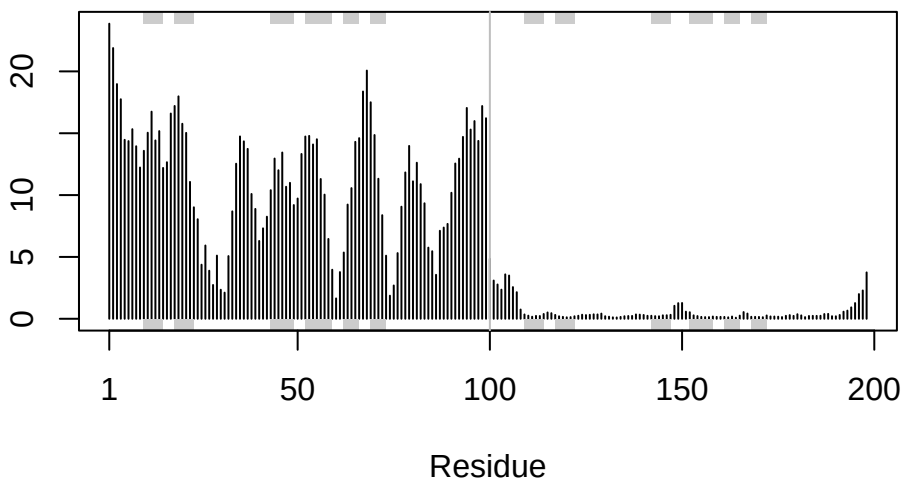
Models 1-3 appear to have high pLDDT scores (most regions are bright red), whereas Models 4 and 5 appear to have more regions of uncertainty (much more white/pink in their structures, reflecting lower pLDDT scores). It also seems that the two sets of models differ greatly in the bottom chain.

RMSF

Calculate the RMSF (measure of conformational variance along the structure).

```
rf <- rmsf(xyz)

# Plot the RMSF values along the (superposed) structure's residues
plotb3(rf, sse=pdb_ref)
abline(v=100, col="gray", ylab="RMSF")
```



It appears that one chain has high variability across the structures (high RMSF values, as shown on left side of plot above), whereas the second has very little variability (low RMSF values, as shown on right side of plot). This is consistent with Mol* observations, as discussed.

Predicted Alignment Error (PAE) for domains

Read in AlphaFold's Predicted Aligned Error (PAE) files, formatted as JSON.

```
library(jsonlite)

# List all PAE JSON files
pae_files <- list.files(path="hivpr_dimer_23119/",
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

# Read all 5 PAE files.
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae2 <- read_json(pae_files[2],simplifyVector = TRUE)
pae3 <- read_json(pae_files[3],simplifyVector = TRUE)
pae4 <- read_json(pae_files[4],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

# Check what attributes are stored in a PAE file.
attributes(pae1)
```

```
$names
[1] "plddt"    "max_pae" "pae"      "ptm"      "iptm"
```

```
# One attribute is the per-residue pLDDT scores
# The values are also in the B-factor column of PDB files

# Print pLDDT scores for the first few residues from 1st PAE file
head(pae1$plddt)
```

```
[1] 91.62 94.06 94.56 93.88 96.12 90.69
```

```
# Determine max PAE value for each model
# i.e. Access the max_pae attribute
pae1$max_pae
```

```
[1] 12.33594
```

```
pae2$max_pae
```

```
[1] 18.67188
```

```
pae3$max_pae
```

```
[1] 14.39062
```

```
pae4$max_pae
```

```
[1] 29.57812
```

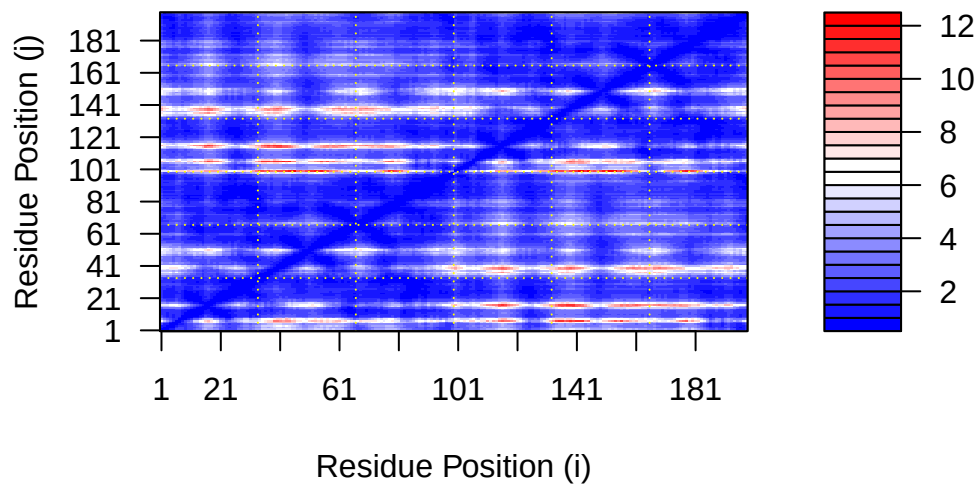
```
pae5$max_pae
```

```
[1] 29.45312
```

The maximum PAE values generally increase going from model 1 to model 5. Model 1 has the best/lowest PAE score of 12.34, whereas Model 5 has a substantially worse/higher PAE score of 29.45. Models 2-4 generally fall between those lower and higher ends of max PAE values (with max PAE scores of 18.67, 14.39, and 29.58, respectively).

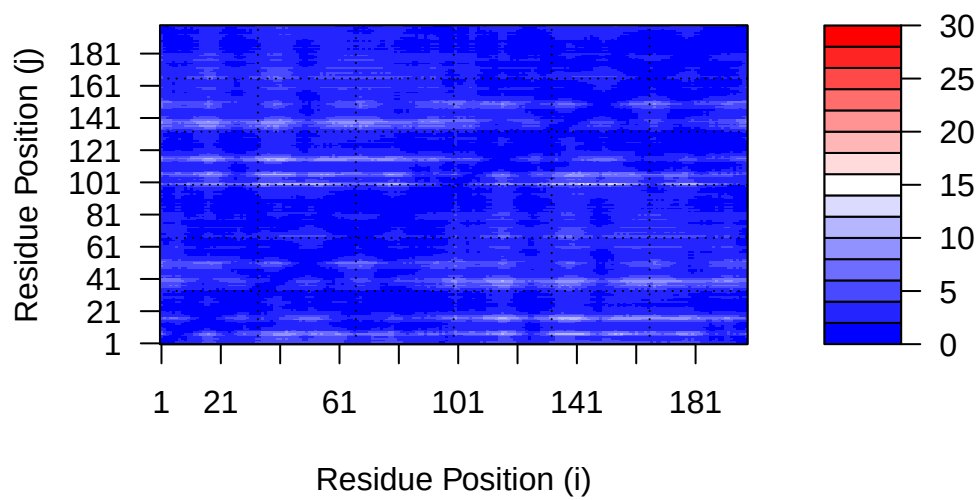
Plot NxN (N = # of residues) PAE scores. You can use ggplot or Bio3D functions, as shown below:

```
# Plot PAE scores for Model 1.  
plot.dmat(pae1$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)")
```

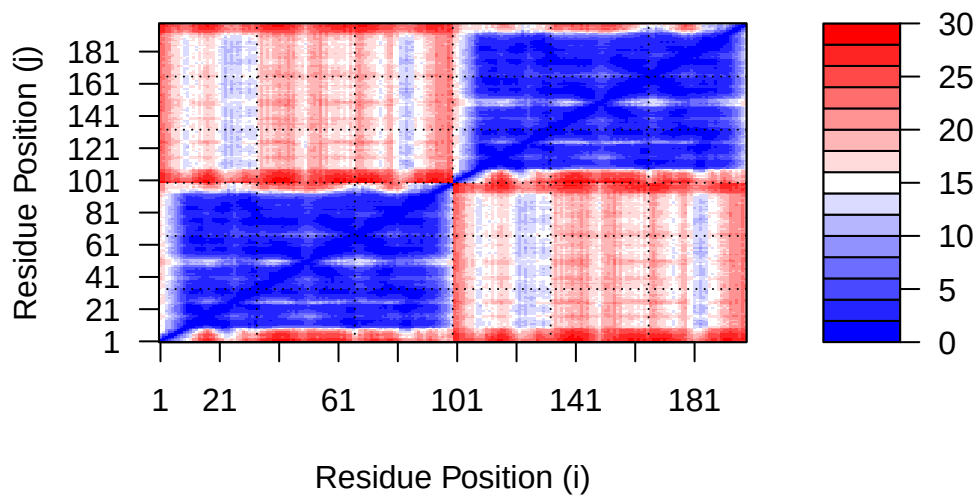


To compare PAE plots for different models, we should ensure they are on the same z range.

```
# Re-plot PAE scores for Model 1, setting (0,30) as z range.
plot.dmat(pae1$paes,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```

```
# Plot PAE scores for Model 5, setting (0,30) as z range.
plot.dmat(pae5$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



Most of Model 1's plot is bright blue, suggesting the model generally has low PAE scores. By contrast, the upper left and lower right quadrants of Model 5's plot (inter-chain residue pairs) are red-dominated, indicating AlphaFold has low confidence in the relative positioning of these residue pairs.

Residue conservation from alignment file

```
# Print out alignment file.
aln_file <- list.files(path="hivpr_dimer_23119/",
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "hivpr_dimer_23119//hivpr_dimer_23119.a3m"
```

```
# Read alignment file.
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
# How many sequences are in the alignment?  
dim(aln$ali)
```

```
[1] 5397 132
```

There are 5397 sequences in the alignment.

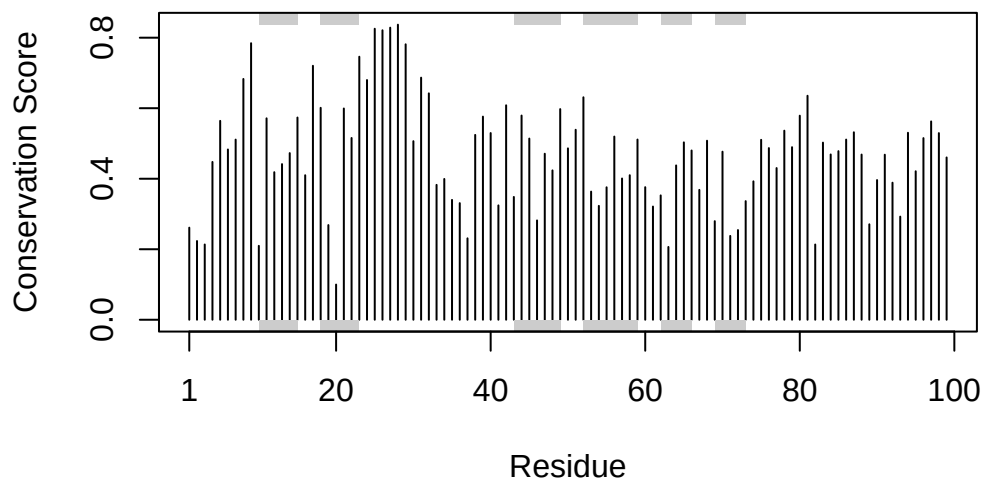
Use the `conserv()` function to score conservation of individual residues in the alignment:

```
sim <- conserv(aln)  
length(sim)
```

```
[1] 132
```

Plot the residue conservation scores:

```
plotb3(sim[1:99],  
       sse=trim.pdb(pdb_ref, chain="A"),  
       ylab="Conservation Score")
```



We can observe from the plot above that residues 25-28 have the highest conservation scores (these residues correspond to a DTGA sequence).

When we generate a consensus sequence using a very high cutoff value, D25, T26, G27, A28 are returned.

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Let's visualize the conserved residues in Mol*.

```
# Write PDB file for model 1, adding Occupancy column (containing conservation scores)
m1.pdb <- read.pdb(pdbfiles[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

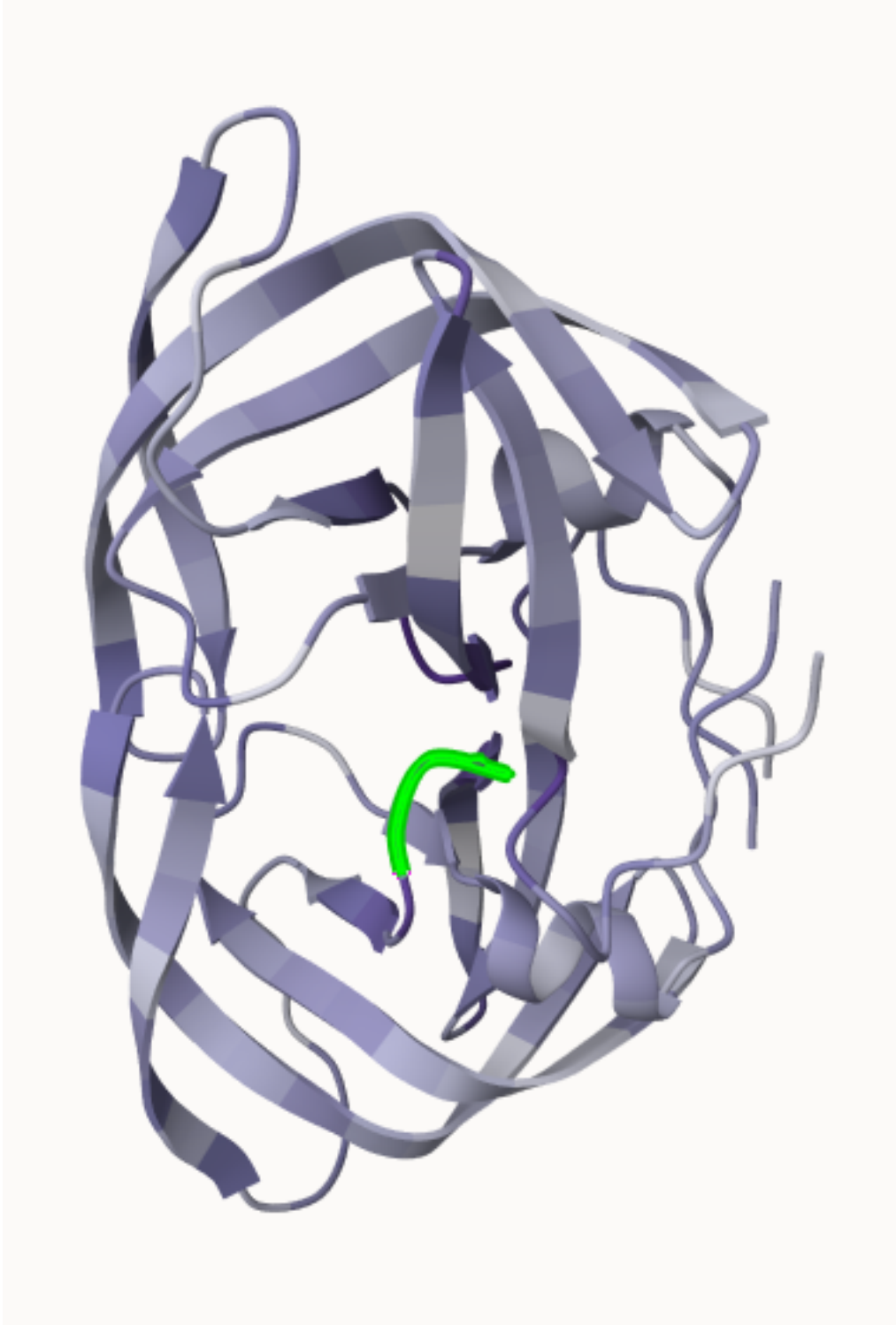


Figure 8: Top ranked AlphaFold model for the HIV protease dimer, colored by residue conservation (i.e. “Occupancy”). More conserved residues are in darker purple. The conserved DTGA motif is selected, in green.

The Mol* visualization, with DTGA highlighted, shows structural conservation at the HIV protease's active site (at the center/interior of the protein). This is a functionally important region of the protease, as it is where substrates bind.